8)

$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ $\quad v \Rightarrow \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ $\quad A = uv^T \Rightarrow \begin{pmatrix} u_1v_1 & u_1v_2 \\ u_2v_1 & u_2v_2 \end{pmatrix}$

u is eigen vector if $\underline{Au = \lambda u}$. where λ is a scalar

$\therefore Au \Rightarrow (uv^T)u = u\underbrace{(v^Tu)}$

$v^Tu = (v_1 \; v_2)\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \Rightarrow (u_1v_1 + u_2v_2) \rightarrow$ Scalar.

$\therefore Au \Rightarrow \lambda_1 u$ where $\lambda_1 \Rightarrow \boxed{v^Tu \Rightarrow u_1v_1 + u_2v_2}$

$\because$ Rank of $A = 1$ $\therefore \lambda_2 = 0$

$\therefore \boxed{\lambda_1 + \lambda_2 \Rightarrow u_1v_1 + u_2v_2}$

Also $\text{trace}(A) \Rightarrow u_1v_1 + u_2v_2$.

$\therefore \boxed{\text{tr}(A) \Rightarrow \lambda_1 + \lambda_2}$